

MINDSPACE

Kanak Gambhirrao | Dhruv Dhapate | Sanket Dongre | Anoop Lanjekar | Riya Bodke | Yash Katolkar

PRESENTED BY:

Group 2

GUIDED BY:

Prof. Vandana Dixit

DATE:

20 February 2023

"You don't have to struggle in silence."

You can be **un-silent**. You can
cure your mind as long as you
open up to somebody about it."

-Demi Lovato



Table of Contents

04

Introduction

06

Overall
Description



08

System
Function



17

External Interface
Requirements

18

Non-Functional
Requirements

20

Other
Requirements

Introduction

- Purpose
- Document Conventions
- Intended Audience and Reading Suggestions
- Project Scope
- Target Audience
- References

Overall Description

- Project Perspective
- Product Features
- User Classes and Characteristics
- Operating Environment
- Design and Implementation Constraints
- User Documentation
- Assumptions and Dependencies

System Function

- Functional Requirements

External Interface Requirements

- User Interface
- Hardware Interfaces
- Software Interfaces
- Communication Interfaces

Non-Functional Requirements

- Performance Requirements
- Safety Requirements
- Security Requirements
- Software Quality Attributes

Other Requirements

- Appendix A: Glossary
 - Appendix B: Analysis Models
 - Appendix C: Issues List
-

Introduction

1.1 Purpose

The document defines and describes the functional and non-functional requirements of social media website, which will exclusively promote solutions on mental health issues. The purpose of this document is that the requirements mentioned in it should be utilized by software developers to implement the system.

1.2 Document Conventions

TERM	DEFINITION
1. FRU	Function Requirement : User Registration & profile settings
2. FRF	Function Requirement : Friends & Suggestions
3. FRP	Function Requirement : Posts
4. FRC	Function Requirement : Chat
5. FRA	Function Requirement : Advertisement

1.3 Intended Audience and Reading Suggestions

This document is intended for a varied set of audiences. It is intended for the design team, the developers, project managers, team leads, supervisors, security analysts, and the testing team to understand the system requirements better.

1.4 Project Scope

Mind Space, a social networking website for mental health, is an online community providing a safe and supportive space for individuals to connect with others experiencing similar mental health issues. The platform provides resources and tools to help individuals manage their mental health, and foster a sense of community and belonging. It is a user-friendly, secure and privacy-conscious website with an easy-to-navigate platform. There will be two types of users who can access the website - a **mentor**, who will be allowed to post content(photos, videos, etc) related to mental care, and a **mentee**, who can browse through the website and avail services, offered by a mentor.

1.5 Target Audience

PEER PRESSURE	STRESS	ISOLATION
		
Arav, 28 years	Gayatri, 16 years	Shekhar, 64 years
• "I wish to kill myself in this environment" - Arav. • Arav is a school-going student, who is suffering from bullying. • He started skipping school as some of his classmates abused him mentally.	• "I hate my job" - Gayatri • Gayatri works in a multinational company and thus have to work for long hours. • She is not happy with her job and is struggling from heavy workload.	• "I wish I could live with my children again" - Shekhar • Being a parent of an NRI, Shekhar lives alone in Bandra. • He is grappling with loneliness syndrome, which leaves him anxious.

Mental health plays a significant role in our day-to-day lives. Our work efficiency mainly depends on how well we mentally are. So, even if you are 16 or 42, the role you play will be affected if you’re mentally ill. Thus our website is open for all age groups. Our goal is to help people like Arav, Gayatri and Shekhar by providing a platform to share their problems.

1.6 References

- <https://chat.openai.com/>
- <https://www.geeksforgeeks.org/>
- <https://www.w3schools.com/>
- <https://stackoverflow.com/>
- https://r.search.yahoo.com/_ylt=AwrKGDDcrFdkulGhwf67HAX;_ylu=Y29sbwNzZzMecG9zAzEEdnRpZAMEc2VjA3Ny/RV=2/RE=1683496284/RO=10/RU=https%3a%2f%2fwww.who.int%2fealth-topics%2fmental-health/RK=2/RS=eL_cNb80J274de19avg8gQOqas-
- https://r.search.yahoo.com/_ylt=Awrx.2cerVdksG4Gtbe7HAX;_ylu=Y29sbwNzZzMecG9zAzEEdnRpZAMEc2VjA3Ny/RV=2/RE=1683496351/RO=10/RU=https%3a%2f%2fwww.questionpro.com%2fsurvey-templates%2fmental-health%2f/RK=2/RS=oY_ru6479nf4jp4MeEznAmfn4fQ-

Overall Description

2.1 Product Perspective

Even if you have good coverage, it can be hard to find a mental health provider. **Mindspace** is a digital platform that provides a space for individuals to connect with others, find support, and access resources related to mental health. Mentors can share photos, videos, and blogs to help Mentees. They can also arrange one-to-one sessions with Mentees and can advertise their therapy sessions. Mental health is equally important as physical health nowadays. Thus our goal is to provide online therapy, support groups, crisis intervention, and self-help resources. These services should be provided by trained mental health professionals and presented in an easy-to-use and accessible format.

2.2 Product Features

Both users can register on the social networking site and log in whenever they wish to do so. They have to set a password to log in to their accounts. Features provided to **Mentee-users**:

- User can create his profile, add, and edit information, set visibility to various profile sections, so his own profile and upload a profile picture to his profile.
- Users can follow Mentor -Users and like or comment on their posts.
- They can chat with Mentors for personal guidance.
- A user can search for specific Mentors and will be shown appropriate results if matching profiles are found.

Features provided to **Mentor-users**:

- User can create his profile, add, and edit information, set visibility to various profile sections, so his own profile and upload a profile picture to his profile.
- They can post photos, videos and blogs related to mental care.
- Users can follow Mentor -Users and like or comment on their posts.
- They can chat with Mentors as well as Mentee Users.
- A user can search for specific Mentors and will be shown appropriate results if matching profiles are found.
- Any user can post status which will be visible to his connections.

2.3 User Classes And Characteristics

Standard users may belong to any demographic group including any gender, nationality and should know how to use a computer browser. The site requires its users to be above 12 years of age. It doesn't require its users to have any specific computer knowledge. User Interface is in English Language so users should have basic English reading knowledge.

2.4 Operating Environment

Hardware Requirements
• Pentium-IV (Processor) • 256 MB RAM • 512 KB Cache Memory • Hard Disk 10 GB • Microsoft Compatible • 101 or more KeyBoard

Software Requirements
• Operating System: Windows/MacOS/Linux • Programming Language: HTML, CSS, JavaScript • Web Development Framework: • Backend : • Database Management System : MongoDB • Content Management System: • APIs: • Testing and Debugging Tools: • Version Control:

2.5 Design And Implementation Constraints

- 1.The website will support the following browsers: Microsoft Edge, Chrome 27.0+ and Firefox 30+
- 2.The website supports English language only.
- 3.The website requires certificates issued by CA to use HTTPS.
- 4.The first version of the website is intended to be viewed best on a desktop browser.

2.6 User Documentation

The social networking website will not provide any documentation. Online tutorial will be served when the user registers to the site.

2.7 Assumptions And Dependencies

- 1.There will be availability of internet via 3G, 4G, 5G or Wi-Fi.
 - 2.The user of the site should be well acquainted with English Language.
 - 3.The user should have a valid email address in order to register on the website.
 - 4.The central server of the system must be able to handle all incoming requests simultaneously.
 - 5.The user should have basic knowledge of computers and the Internet.
-

System Function

3.3 Functional Requirements



For A **Mentee**

1. User Registration

ID - FRU 1

Description -

The user will create an account if there is no existing account associated with the user's email id. User should enter account details such as **name**, **email-id**, **contact number**, **birth date** and **password**. All these are required fields and account creation cannot proceed unless these are filled. The user is registered after this information is validated and the user **agrees** with the **terms and conditions**. The user must be 12 years old or more. The **contact number** should be a **10-digit numeric field**. **First name** and **last name** should not exceed **20 characters** each. **Password** should be at **least 8 characters** long, at **most 15 characters** including at least **one capital** and **one small letter, a special character** and **a number**.

Dependency - None

2. Login

ID - FRU 2

Description - The user **must be registered** in order to log in. **Input** will be the **email id** and **password** for login. The **hash of the password should match the hash of the password stored in the database**. If **three attempts** of the password are **wrong** then a **Captcha** should appear. If **more than 8 attempts** go **wrong**, then the **account will be locked** and an **e-mail** will be sent to the user notifying about the wrong password attempts. To unlock the account, the user must click on the **unlock link** sent to him in the email. If the user clicks on **forgot password** then the **reset password link** should be emailed to the user. Input to each field must be validated.

Dependency - FRU 1

3. Add/Edit Issue details

ID - FRU 3

Description - A part of building a user profile. The user should be able to **add the Mental issue details**. This field will be **only visible to the user and his Mentors**. These fields can remain empty. The user must be **logged in** and his account should be marked as **verified** to be able to add education. Input to each field must be validated.

Dependency - FRU 2

4. Add/Edit Personal information

ID- FRU 4

Description - A part of building a user profile. The user should be able to **add personal details** like **language, gender**, about the user and his **interests**. These fields **can remain empty**. The user must be **logged in** and his account should be marked as **verified** to be able to add personal information. Input to each field must be validated.

Dependency- FRU 2

5. Add/Edit account information

ID – FRU 5

Description - A part of building a user profile. The user should be able to add the account details like **name, birth date, email-id, phone number** and **address**. **Name, birth date** and **email-id cannot be empty** while the other two can remain empty. The user must be logged in. Input to each field must be validated.

Dependency -FRU 2

6. Upload Profile Picture

ID – FRU 6

Description - The user should be able to **upload** the **profile picture**. The **size** of the profile picture should be **at least 160px* 160px**. Users may leave this field empty. The **profile image** should be **visible to all**.

Dependency-None

7. Set account visibility settings

ID- FRU 7

Description -The user should be able to make **account details private** or **public** according to his wish. By default, the account details will be visible to other Mentor users as well as Mentee users.

Dependency -None

8. Set personal information visibility

ID- FRU 8

Description- The user should be able to make **personal information private** or **public** according to his wish. By default personal information will be visible to both Mentor users as well as Mentee users.

Dependency –None

9. Show Profile

ID – FRU 9

Description - Any logged-in user should be **able to see any Mentor or Mentee profile**.

Dependency - FRU 1, FRU 2

10. Delete Account

ID - FRU 10

Description - The user may **delete his account permanently** depending on his wish. User will be **removed** from the **mentee list** of his **respective mentor** once deleted. User information will be cleared.

Dependency - FRU 1, FRU 2

11. Follow/Unfollow A Mentor

ID – FRF 1

Description - A registered Mentee user should be able to **send follow requests** to Mentors. Users can **search** for **specific renowned Mentors** among the registered users and then send a follow request to him/her. User can also unfollow them.

Dependency - FRU 1

12. List Mentors

ID - FRF 2

Description - Whenever the user is on any person's profile page, he should have the option to view the currently open user's **Mentor list** on a page. A User can have many Mentors and the maximum number of Mentors shown at a time will be 15, and paging will be implemented to show more Mentors if the user has more than 15 Mentors.

Dependency- FRU 2, FRU 1, FRF 1

13. Search User

ID- FRF 3

Description - A User should be able to **search for Mentors or Mentees** who may be registered on the site. The user will put the name of the Mentor or Mentee and click search after which the search results will display all the people registered on the site by the searched name. When the search results are displayed the **maximum number of results** shown at a time will be **20**, and paging will be implemented to show more results.

Dependency - FRU 3

14. Invite a Friend

ID- FRF 4

Description - The user should be able to **invite more friends** to join the social network. Users will be given an option to enter the person's email address to invite. Users will be able to invite one friend at a time.

Dependency - FRU 3, FRU 1

15. Post Visibility

ID - FRP 1

Description- *Posts* posted by any Mentor should be **visible to all registered Mentee users**. If they find a post offensive, scam, hate speech or violent, they can report it by clicking the **REPORT button** on the post.

Dependency- FRU 2

16. Like

ID- FRP 2

Description - A user can **like a post** that has been posted to show support to Mentors.

Dependency - FRU 2, FRP 1

17. Comment

ID- FRP 3

Description -For each post visible to the user he has the ability to **comment** on it to his opinion. The comment can contain only text and cannot exceed **500 characters** in length. To comment, the user can see a **text area under the post**, where he can enter the text.

Dependency- FRU 3, FRP 1

18. Delete Comment

ID- FRP 4

Description - A user can **delete a comment** written by him. For deleting a post that he is authorized to delete, a user can see a **delete link** near the comment.

Dependency- FRU 2, FRP 3

19. Chat

ID - FRC1

Description - The user should be able to initiate a **chat with any Mentor** who is online. The Mentor user must seamlessly understand that the chat has been initiated and the chat message must be forwarded to the intended Mentor. Users should be able to **receive messages from any Mentor** who is online. Thus the receiving entity can send a message back to the originator, thus, completing a two-way conversation. The message cannot be greater than **250 characters** in length.

Dependency - FRU2

20. Advertisement

ID-FRA1

Description-User should be able to **view the ads** displayed alongside his wall.

Dependency-FRU2

For A **Mentor**

1. User Registration

ID - FRU 1

Description - The user will create an account if there is no existing account associated with the user's email id. Users should enter account details such as **name**, **email-id**, **contact number**, **birth date** and **password**. All these are required fields and account creation cannot proceed unless these are filled. Mentors have to **fill out a form** to **verify** their **qualifications**. The user is registered after this information is validated and the user **agrees** with the **terms and conditions**. The **contact number** should be a **10-digit numeric field**. **First name** and **last name** should not exceed **20 characters** each. **Password** should be at **least 8 characters** long, at **most 15 characters** including at least **one capital** and **one small letter**, **a special character** and **a number**.

Dependency - None

2. Login

ID - FRU 2

Description - The user **must be registered** in order to log in. **Input** will be the **email id** and **password** for login. The **hash of the password should match the hash of the password stored in the database**. If **three attempts** of the password are **wrong** then a **Captcha** should appear. If **more than 8 attempts** go **wrong**, then the **account will be locked** and an **e-mail** will be sent to the user notifying about the wrong password attempts. To unlock the account, the user must click on the **unlock link** sent to him in the email. If the user clicks on **forgot password** then the **reset password link** should be emailed to the user. Input to each field must be validated.

Dependency - FRU 1

3. Add/Edit Qualification

ID - FRU 3

Description - A part of building a user profile. The user should be able to **add or edit** the **qualification details**. These fields can remain empty. The user must be **logged in** and his account should be marked as **verified** to be able to add education. Input to each field must be validated.

Dependency - FRU 2

4. Add/Edit Personal information

ID- FRU4

Description - A part of building a user profile. The user should be able to **add or edit** **personal details** like **language**, **gender**, about the user and his interests. These fields can remain empty. The user must be logged in and his account should be marked as verified to be able to add personal information. Input to each field must be validated.

Dependency- FRU2

5. Add/Edit account information

ID – FRU 4

Description - A part of building a user profile. The user should be able to **add/edit** the **account details** like **name, birth date, email-id, phone number** and **address**. Name, birth date and email-id cannot be empty while the other two can remain empty. The user must be logged in. Input to each field must be validated.

Dependency -FRU 2

6. Upload Profile Picture

ID – FRU 5

Description - The user should be able to upload the profile picture. The **size** of the profile picture should be **at least 160px*160px**. Users may leave this field empty. The profile image should be visible to all.

Dependency-None

7. Set account visibility settings

ID- FRU 7

Description -The user should be able to make **account details private** or **public** according to his wish. By default, the account details will be visible to the public.

Dependency -None

8. Set personal information visibility

ID- FRU 8

Description- The user should be able to make **personal information private** or **public** according to his wish. By default personal information will be visible to the public.

Dependency –None

9. Show Profile

ID – FRU 9

Description - Any logged-in user should be able to see any **Mentee** or **Mentor** user **profile**.

Dependency -FRU 1, FRU 2

10. Delete Account

ID - FRU 10

Description - The user may **delete his account permanently** depending on his wish. The user will be removed from his **Mentor** or **Mentee** list once deleted. User information will be cleared.

Dependency - FRU 1, FRU 2

11. Follow A Mentor

ID – FRF 1

Description - A registered user should be able to **send follow requests** to other users. Users can send follow requests to already registered users. They can search for specific Mentors or Mentees among the registered users and then send a friend request to him/her.

Dependency - FRU 1

12. Accept/Ignore/Delete Mentee Follow Request

ID – FRF 2

Description - If a user gets a **follow request from** another **Mentee or Mentor user**, then the user receiving the request should have the option to **accept, ignore or delete** the follow request. If the user selects to **accept the follow request**, the **requester** is **added to the list** of the acceptor. If the user selected to **ignore the request**, the **sender** continues to see the **status** of the request as **“request sent”** and the **request is removed from** the queue of the **request receiver**. If a user selects to **delete** a specific Mentor or Mentee from his list, the Mentor or Mentee he selects gets removed from his list.

Dependency - FRF 1

13. Unfollow A Mentor

ID- FRF 3

Description – Any registered user should be able to **un-follow** any of the existing **Mentors**.

Dependency - FRU 1, FRF 1

14. Block/Report A Mentee

ID- FRF 4

Description- Any registered user should be block or report any of the existing Mentors as well as Mentees.

Dependency- FRU 1, FRF 1

15. Invite a Friend

ID- FRF 5

Description - The user should be able to **invite more friends** to join the social network. Users will be given the option to enter the person's **email** address to invite. Users will be able to invite **one friend** at a time.

Dependency - FRU 3, FRU 1

16. List Mentee

ID - FRF 6

Description - Whenever the user is on any Mentor's profile page, he should have the option to view the current user's **Mentee-list on** a page. Users can have many mentees and the maximum number of mentees **shown at a time** will be **20**, and paging will be implemented to show more Mentees if the user has more than 20 Mentees.

Dependency- FRU 2, FRU 1, FRF 1

17.Search User

ID- FRF 7

Description - The user should be able to search for **specific Mentors for Mentees** who may be registered on the site. The user will put the name of the Mentor or Mentee and click search after which the search results will display all the people registered on the site by the searched name. When the search results are displayed the maximum number of results shown at a time will be **20**, and paging will be implemented to show more results.

Dependency - FRU 3

18. Post

ID - FRP 1

Description- A registered **Mentor user** can **share posts** with his Mentees. The post can only be plain English text with maximum **500 characters**. The user can enter the post in the designated text area and clicking **POST** button, should be visible to the user and his Mentees.

Dependency- FRU 2

19. Remove Post

ID- FRP 2

Description - A Mentor user can **remove a post** that has been posted. After deletion, the post should not be visible on the user's; or his Mentee-user's profiles. Deletion of a post should also remove all the comments associated with that post.

Dependency - FRU 2, FRP 1

20. Comment

ID- FRP 3

Description -For each post visible to the user he has the **ability to comment** on it to his opinion. The comment can contain only text and cannot exceed **500 characters** in length. To comment, the user can see a text area under the post, where he can enter the text.

Dependency- FRU 3, FRP 1

21. Delete Comment

ID- FRP 4

Description- A user can **delete a comment** written by him or a comment on his posts. For deleting a post that he is authorized to delete, a user can see a **delete link** near the comment.

Dependency- FRU 2, FRP 3

22. Chat

ID - FRC 1

Description - The user should be able to initiate a chat with any Mentor or Mentee user in the list who is online. The other user must seamlessly understand that the chat has been initiated and the chat message must be forwarded to the intended user. Users should be able to receive messages from any user in the list who is online. Thus the receiving entity can send a message back to the originator, thus, completing a two-way conversation. The message cannot be greater than **250 characters** in length.

Dependency - FRU 2

23. Advertisement

ID- FRA1

Description- The user should be able to **view the ads** displayed alongside his wall. A user should also be able to **post an ad**.

Dependency- FRU2

External Interface Requirements

4.1 User Interfaces

MindSpace has two user interfaces -

- **Mentee-user Interface :**

It shall have a login page for users to login into the site. For those who are first-time users, they will have to fill up the registration details. After logging into the system, the user will be asked to fill out a questionnaire about his issues and will suggest suitable Mentor profiles. After logging into the system, the user will be shown his profile and the wall containing the posts shared by the Mentors they follow. The user is given options to log out, search Mentors, follow Mentors, unfollow Mentors, message Mentors and change any of his personal settings on his home page.

- **Mentor-user Interface :**

It shall have a login page for users to login into the site. Those who are first-time users will have to verify their qualifications through a form. After completing the registration details, the user will get verified as a MindSpace Mentor. After logging into the system, the user will be shown his profile and the wall containing posts shared by other Mentors they follow. The user is given options to log out, search people and change any of his personal settings on his home page.

4.2 Hardware Interfaces

No hardware is required.

4.3 Software Interfaces

- **E-mail Interface**

This interface uses the SMTP/POP services provided by a third party to send emails to the required user. This service will notify users of required actions through a conventional medium.

- **Captcha Service Interface**

Captcha Services will be used for human Identification

4.4 Communication Interfaces

The social networking website is based on HTTPS. Moreover, TCP/IP is used for chatting functionality.

Non-Functional Requirements

5.1 Performance Requirements

- Scalability

The system should be able to handle a large number of users. For e.g. handling around a hundred users at the same time.

- Speed

The application should be fast. It should not slow down with an increase in the number of users.

Search functionality should be fast to enable a better end-user experience.

The system should be quick enough to be able to respond to the user's actions within a short period of time.

E.g. the search user functionality of the site should perform a quick search among the users on the database.

5.2 Safety Requirements

N/A

5.3 Security Requirements

- During user registration, the given email address is validated.
- The password should be at least 8 characters, containing at least a small character and one capital, a number and a special character.
- Password is stored as a hash value in database.
- We are transferring all data via HTTPS i.e. via SSL so that the data is encrypted during the transit. Thus safeguarding the user information.
- Re-Captcha service is used during registration for human identification.

5.4 Software Quality Attributes

- **Usability**

The user interface should be simple and clear to be understood by any user.

- **Availability**

The system should be available at all times. It should be ensured that there should be minimal or no downtime to ensure a better user experience. The system should be reliable. It should yield correct results if a user performs searches for a person. Also, if the user sends a message, the system should ensure that the correct message is delivered to the correct destination without any loss or change in content.

- **Testability**

The application should be testable. A separate test environment should be set up where testers and the Quality Assurance engineers can test the application for bugs and/or incomplete or missed requirements.

- **Maintainability**

The system should be developed in such a way that it is extensible. It should be easy to incorporate new feature requirements or accommodate a change in the existing requirements.

Other Requirements

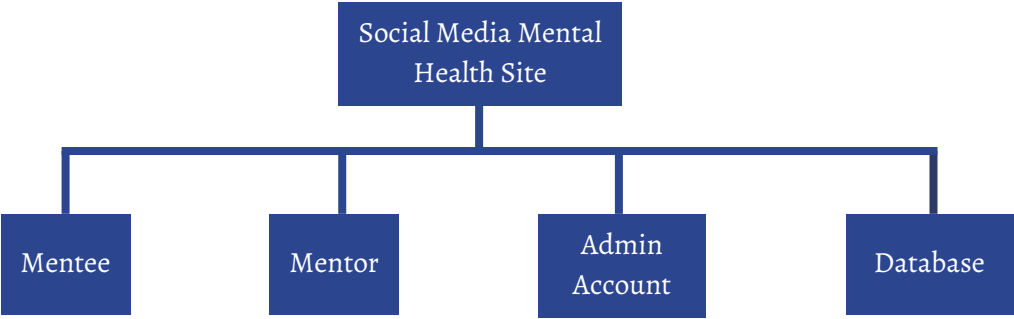
6.1 Appendix A: Glossary

- **Mentee**: A user who can browse the website
 - **Mentee List**: A List of Mentee followers(Only available for Mentor-users)
 - **Mentor**: A verified influencer who will post content related to mental help
 - **Mentor List**: A List of Mentors a user follows
-

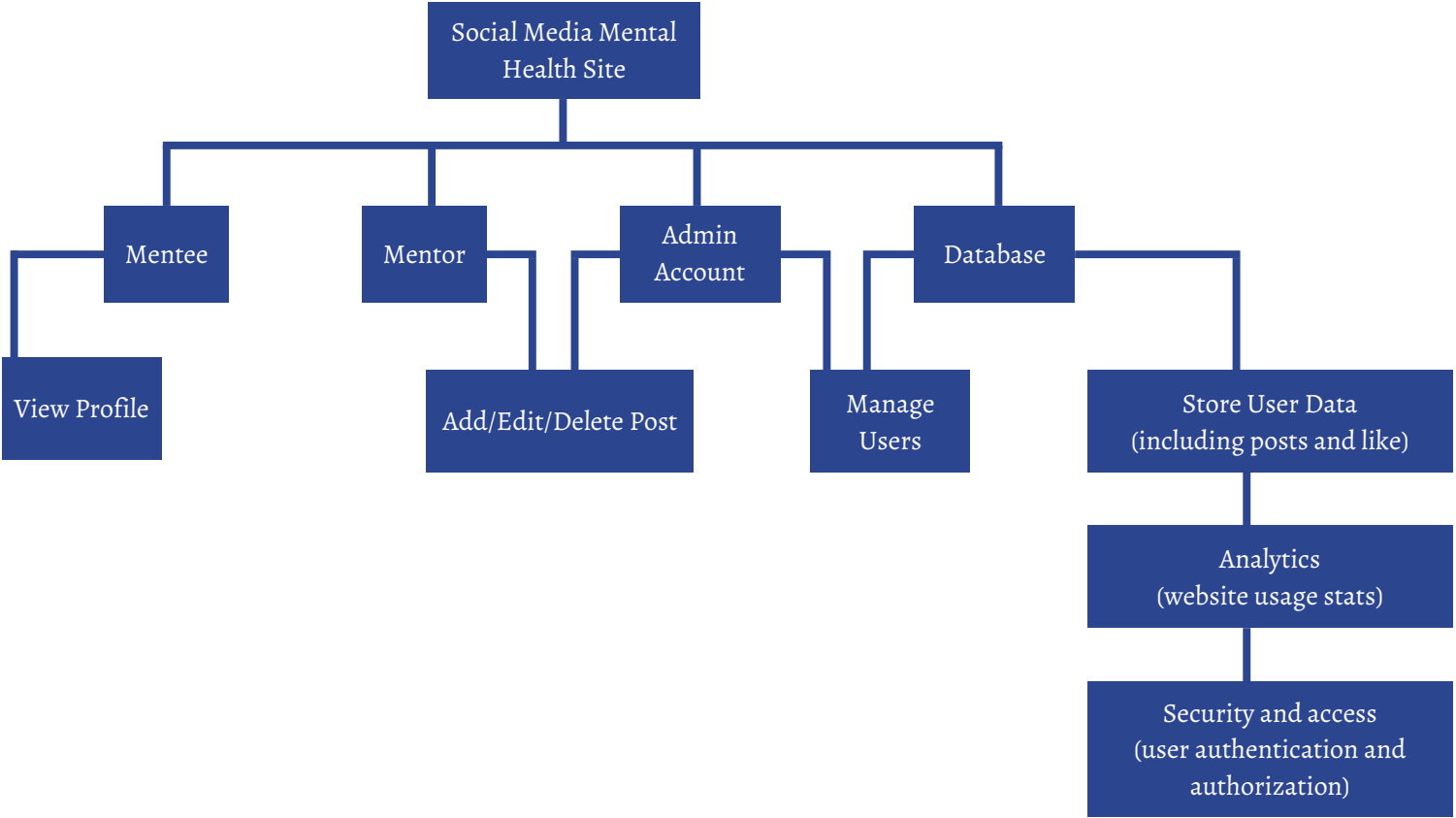
6.2 Appendix B: Analysis Model

- DFD

Level 0 :



Level 1 :



6.3 Appendix C: Issues List

MISSION: Provide a platform for seeking mental help	VISION: Making it easy to find a mental help provider	VALUE PROPOSITION:
--	--	--------------------